

AKSHAY S

Alappuzha, Kerala, India

+91 8129163092 ✉ akshayaththam@gmail.com [linkedin.com/in/akshay-s31](https://www.linkedin.com/in/akshay-s31) github.com/Akshay-S-31

Education

TKM College Of Engineering

Bachelor of Technology in Computer Science(Hons.)

Sep. 2023 – May 2027

Kollam, Kerala

Technical Skills

Languages: Java, Python, HTML/CSS, JavaScript, SQL

Technologies/Frameworks: Spring Boot, Node.js, Express.js, OpenCV, Scikit-learn, PyTorch

Developer Tools: Git, GitHub, AWS, Linux

Databases: MySQL, PostgreSQL, MongoDB

Experience

Susaanoo Systems

May 2026 – July 2026

ML Intern

Remote

- Architected a **Hybrid Spatial-Temporal** Deep Learning Engine combining a Variable Selection Network (VSN), multi-head Graph Neural Networks (GNNs), and a **Mamba-3 MIMO State Space Model** (SSM) to capture joint cross-asset dependencies across BTC, ETH, and SOL without sequence flattening.
- Formulated a **Cost-Aware Dual-Head Loss Objective** enforcing monotonic quantile intervals while concurrently classifying 3-class directional targets bounded by a 0.1% exchange fee threshold across 15m, 1h, and 4h horizons.
- Implemented strict 5-Fold Chronological Walk-Forward Validation on order-book microstructure data, outperforming Gradient Boosted Tree baselines by achieving up to **0.668 out-of-sample ROC-AUC** and **48.8% actionable win-rate** on non-neutral trading signals.

Projects

URL Shortener | Java, Spring Boot, Spring Data JPA, H2, Maven

[GitHub](#)

- Built a RESTful URL shortening service with **Spring Boot**, implementing a layered **MVC architecture** with service interfaces for dependency inversion.
- Designed a collision-free short code generator using **Base62 encoding** on auto-incremented database IDs, eliminating the need for collision-check loops.
- Implemented persistent storage with **Spring Data JPA** and **H2** (file mode), with indexed lookups on the short code column to optimise the read-heavy redirect path.

Signora – AI-Based Sign Language Learning Platform | Node.js, Express.js, MediaPipe

[GitHub](#)

- Designed and built **RESTful APIs** for lesson evaluation and user progress tracking, with MongoDB for scalable performance data storage.
- Developed a full-stack **MERN** application with real-time hand landmark tracking using **MediaPipe** and **OpenCV** via live webcam feed.
- Built a **Random Forest** classification pipeline to evaluate user-performed gestures and deliver instant corrective feedback in React.

LUCID – Semi-Supervised Deep Learning Framework for Image Dehazing | Python, PyTorch

[GitHub](#)

- Architected a **PyTorch** deep learning framework (**LUCID**) for real-world computer vision tasks, specializing in image dehazing and restoration to significantly enhance visual clarity on complex datasets (**RTTS**).
- Resolved a critical mathematical variance deadlock in a semi-supervised architecture by engineering a novel deterministic **physical drift error**, improving the model's ability to learn from unannotated data.
- Optimized loss functions for hallucination suppression by implementing dynamic masking, achieving a +2.88 dB **PSNR** improvement and an industry-leading **FADE** score of 0.7239 over baseline models.

Relevant Coursework

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|-----------------------|-----------------------|---------------------|-------------------|
| • Data Structures | • Database Management | • Operating Systems | • Object Oriented |
| • Algorithms Analysis | • Machine Learning | • Computer Networks | Programming |

Certifications

- AWS Cloud Technical Essentials [🔗](#)
- Programming, Data Structures and Algorithms – NPTEL (IIT Madras) [🔗](#)
- IBM SkillsBuild: From Learner to Builder – AI Agent Architect [🔗](#)